# FY-4A + NWP Paired Sample Database

**Target Region:** East China (20°–45°N, 100°–135°E) **Observation:** FY-4A AGRI L1 Radiance **Background:** ERA5 Reanalysis

**The Task:** Build a paired sample database in which each sample links a numerical model state to a FY-4A satellite image, to be used for studying how information in satellite imagery can be assimilated into a numerical weather prediction model.

## 1. What would you pair with what? Why those two things?

I would pair **top-of-atmosphere (TOA) brightness temperatures from FY-4A AGRI infrared (IR) and water vapor channels** with the **3D thermodynamic, kinematic, and hydrometeor state of the NWP model** across standard pressure levels (1000 to 250 hPa), supplemented by key 2D surface parameters.

**Satellite Observations:**

- `Channel 9 (6.25 μm)`: High-sensitivity absorption channel sensing upper-to-mid tropospheric water vapor.
- `Channel 10 (7.1 μm)`: Mid-to-lower tropospheric moisture; paired with C09 as a differential index (`C09 - C10`), it retrieves vertical moisture lapse rates.
- `Window Channels C12 (10.7 μm) & C13 (12.0 μm)`: Clean and split-window channels for cloud-top and skin temperature; the split difference (`C12 - C13`) diagnoses boundary-layer moisture, aerosols, and cloud optical depth.
- `CO₂ Slicing (C14, 13.5 μm)`: Resolves vertical atmospheric transmission for accurate Cloud Top Height (CTH) and semi-transparent cirrus detection.
- `Temporal Radiance Tendency (ΔT_b / 15 min)`: Differencing consecutive 15-minute scans captures rapid cloud-top cooling (< -10 K / 15 min), the signature of convective storm initiation.
- `Viewing & Solar Geometry`: Satellite zenith/azimuth angles (airmass optical path length) and solar zenith angle.

**NWP Model State:** In addition to thermodynamic variables (`T, q, u, v, z`), modern all-sky assimilation requires **cloud microphysics**:

- Cloud Liquid Water (`q_l`) & Cloud Ice Water (`q_i`) per pressure level (kg/kg).
- Rain and Snow Water Content (`q_r, q_s`) and Fractional Cloud Cover (`cc`).
- Column integrals and surface conditions: Surface Pressure (`P_s`), 2m Temperature (`T_2m`), Skin Temperature (`T_skin`), Total Column Water Vapor (TCWV), Total Column Cloud Liquid (TCLW), Land-Sea Mask (`lsm`), and Surface Orography (DEM).

**Why Pair These Variables:** In physical data assimilation, observed radiances are non-linear vertical integrals of temperature, moisture, and hydrometeor profiles governed by Radiative Transfer Models (RTM). Pairing multi-spectral IR/WV radiances directly targets the prognostic state variables updated by variational (3D/4D-Var) or ensemble Kalman filter schemes. Furthermore, IR/WV channels operate continuously 24/7 independent of daylight. Including cloud hydrometeors transitions the database from restrictive clear-sky assumptions to full all-sky assimilation, allowing AI models to learn direct mappings from satellite radiances to cloud water and precipitation.

## 2. How would you decide that a pair is correctly matched? What criterion would you use, and how large an error would you consider acceptable?

A pair is declared valid if and only if it satisfies three rigorous criteria across time, space, and physical integrity:

- **Temporal Matching:** The FY-4A scan timestamp must align with the NWP model validity time within **≤ 15 minutes** (hard ceiling **< 30 minutes**). Because convective clouds, frontal moisture boundaries, and vertical updrafts evolve on sub-hourly timescales, larger discrepancies produce unphysical representativeness error. Matching 5–15 minute satellite scans to hourly NWP fields yields an expected average offset of **~11–15 minutes** (11.85 min achieved in our prototype), refinable via First Guess at Appropriate Time (FGAT) or 4D temporal interpolation.
- **Spatial Registration & Resolution Alignment:** The native FY-4A geostationary projection (~4 km nadir) is regridded to the NWP regular lat-lon grid (**0.25° × 0.25° ≈ 28 km**) using area-weighted spatial aggregation. The acceptable spatial registration error is **< 0.125° (< 14 km)**—half a model grid cell.
- **Physical Bounds & Quality Control:** Samples are rejected if missing/corrupted pixels exceed **5%** within the analysis domain, or if calibrated brightness temperatures fall outside the physical atmospheric window of **[170 K, 340 K]**. NWP missing data must not exceed **5%**.

### 3. What would a single record in this database contain? List the fields you would define.

Each database record is encapsulated in a CF-compliant NetCDF4 or Zarr chunked file (e.g. `[sample_id].nc`), partitioned into four coordinated groups:

| Data Group | Fields & Defined Parameters | Dimensions | Units / Notes |
|---|---|---|---|
| **1. Metadata & Coordinates** | `sample_id`, `obs_time_utc`, `model_time_utc`,<br>`time_diff_minutes` (Δt),<br>`latitude` (20.0° to 45.0°N),<br>`longitude` (100.0° to 135.0°E) | Scalar / 1D<br>[101]<br>[141] | ISO8601 UTC<br>Minutes<br>0.25° Common Grid |
| **2. Observation State (y)** | Calibrated Brightness Temp (`C09, C10, C12, C13, C14`),<br>Spectral Differences (`C12 - C13`, `C09 - C10`),<br>Temporal Radiance Tendency (ΔT_b / 15 min),<br>Satellite Viewing Angles (Zenith θ_sat, Azimuth ϕ_sat),<br>Solar Zenith Angle (θ_sol), Pixel Quality Flags | [101 × 141 × 5]<br>[101 × 141 × 2]<br>[101 × 141 × 3]<br>[101 × 141 × 2]<br>[101 × 141] | Kelvin (K)<br>Kelvin (K)<br>K / 15 min<br>Degrees (°)<br>Quality bitmask |
| **3. Model Background (x)** | 3D Thermodynamic & Kinematic Fields:<br>- Temperature (T), Specific Humidity (q)<br>- Horizontal Winds (`u, v`), Geopotential (z)<br>3D Cloud Microphysics (All-Sky):<br>- Cloud Liquid (`q_l`), Cloud Ice (`q_i`)<br>- Rain/Snow Water (`q_r, q_s`), Cloud Fraction (cc)<br>2D Surface & Column Fields:<br>- `P_s`, `T_2m`, `T_skin`, `TCWV`, `TCLW`, DEM, LSM | 7 levels × [101 × 141]<br>(1000, 925, 850, 700, 500, 300, 250 hPa)<br><br>7 levels × [101 × 141]<br><br>[101 × 141] | K, kg/kg<br>m/s, m²/s²<br><br>kg/kg<br>kg/kg, 0–1<br><br>Pa, K, K, kg/m², m, 0–1 |
| **4. QC & Partitioning** | `missing_fraction_sat`,<br>`missing_fraction_nwp`,<br>`overall_valid` (Boolean),<br>`split_partition` (train / val / test) | Scalar<br>Scalar<br>Categorical | Percentage (0.0–1.0)<br>True / False<br>Chronological Split |

### 4. In your view, which step of this task is the hardest, and why?

In my view, the hardest step is **reconciling spatial representativeness and cloud parallax displacement** when linking geostationary satellite radiances to grid-averaged NWP model columns.

**1. The Parallax Geometry Problem:** Because FY-4A is stationed at 104.7°E above the equator, line-of-sight paths intersect mid-latitude domains (East China, 20°–45°N) at oblique zenith angles of 40°–60°. High convective cloud towers reaching 12–16 km altitude are displaced laterally on the projection plane by **15 to 40 km** away from their true ground nadir position. If uncorrected, cold cloud-top radiances map to the wrong vertical atmospheric column in the NWP state, injecting artificial dipole innovation errors into the assimilation system.

**2. Non-Linear Multi-Scale Representativeness:** Satellite infrared radiometers sample continuous sub-grid cloud heterogeneity at ~4 km, whereas NWP models represent discrete volumetric averages at ~28 km (0.25°). Radiative transfer through a partly cloudy grid cell is fundamentally non-linear: the average brightness temperature does not equal the radiative transfer of the average atmospheric state ($E[RTM(x)] \neq RTM(E[x])$). Aggregating satellite radiances to the model grid without smoothing out sharp localized physical gradients (such as convective cores and drylines) is the primary physical and computational bottleneck.

**Prototype Pipeline Validation:** To test this framework in practice, I built a modular Python pipeline and validated it on real FY-4A AGRI Level-1 HDF observations and matching ERA5 atmospheric states over East China (20°–45°N, 100°–135°E). The pipeline successfully calibrated raw counts using instrument Look-Up Tables, performed geostationary inverse coordinate remapping, and generated a clean dataset of **17 paired samples** (100% valid, 0.0% missing data in domain, mean Δt = 11.85 min) partitioned chronologically into training, validation, and test sets.